Manifold LLC

Independent AI/ML advice for Intelligence Community program offices

Manifold LLC is a Virginia small business whose people spent about ten years each inside IC acquisition offices and research programs as trusted agents and SETA engineers, on the government's side of the table. We help program managers decide what to buy, what to build, and how to get it running on the high side. When the answer is build, we build; never for a sponsor we also advise.

CORE COMPETENCIES

Strategy and roadmaps

Foundation-model strategy, roadmaps, build-versus-buy analysis, technology forecasts. Technical SME on competitive acquisitions and BAA panels; AI/ML training for sponsor staff.

Acquisition and compute

Technical requirements, RFP and SOW language, source-selection criteria, and vendor evaluation. GPU and TPU cluster sizing and cost models across AWS GovCloud, Azure Government, and Google Cloud's assured regions, or on-premises hardware.

Program assessment

Independent technical evaluation of AI/ML programs, maturity and risk assessment, milestone and design reviews, sponsor-facing metrics.

Secure generative AI

Reference architectures for CUI at IL4 and IL5, retrieval-augmented generation design and delivery, model and vendor selection in government cloud regions, evaluation and bias audits.

Applied research and development

Detectors and trackers over satellite imagery, foundation models pretrained on sponsor data with scaling studies, CPU, GPU and TPU implementations priced per run. Evaluation protocol first; negative results reported.

PAST PERFORMANCE · THE FOUNDERS, BEFORE MANIFOLD

NAVY UARC · GENERATIVE AI
Designed and delivered a production retrieval-augmented generation system for CUI on Amazon Bedrock in an IL4 AWS GovCloud environment, with LDAP integration.

GEOSPATIAL FOUNDATION MODELS
Architected the modeling and training effort for foundation models over overhead data, working directly with Google and NVIDIA engineering teams on pipelines, model design and compute planning.

INTELLIGENCE COMMUNITY ACQUISITION
About ten years inside IC acquisition offices and research programs: requirements, source-selection and BAA evaluation, AI/ML training design, leadership briefings.

DIFFERENTIATORS

- Everyone at the firm holds TS/SCI with CI polygraph
- Senior practitioners deliver the work themselves
- Hands-on with both acquisition and model training
- Delivered production generative AI at IL4 on GovCloud
- Advise or build, never both for one client; no vendor fees

COMPANY DATA

LEGAL NAME	Manifold LLC (Virginia, formed 2026)
LOCATION	Loudoun County, Virginia
SIZE	Small business
UEI	C6P2G6PQLEB8
CAGE	23X60
NAICS	541715, 541512, 541611, 541690
CLEARANCES	TS/SCI with CI polygraph, everyone at the firm
CLASSIFIED WORK	On customer premises
CONTRACT VEHICLES	None yet. Direct award, subcontract or teaming
PRINCIPALS	Paul G. Singerman, Ph.D.; John "Luke" Och

POINT OF CONTACT

CONTACT	Paul G. Singerman, Principal
EMAIL	info@manifold-llc.com
WEB	manifold-llc.com

RESEARCH

MARITIME DOMAIN AWARENESS · RESEARCH

Built and measured a tracker that re-finds ships that have switched off AIS from one satellite pass to the next, over the waters around Denmark and southern Norway, using public radar, optical and night-light sensors (Sentinel-1, NISAR, Sentinel-2, Landsat, VIIRS), a motion model learned from historical AIS, and scoring against withheld Danish and Norwegian AIS. No published system was found that measures this problem; the results would be a first public baseline once published. Unpublished draft.

SYNTHETIC APERTURE RADAR · OPEN-SOURCE SOFTWARE

FastSAR, a library that forms SAR images on CPUs, NVIDIA GPUs and Google TPUs through one Python interface (exact and factorized backprojection, polar format; CPHD, SICD, Sentinel-1 and NISAR inputs). Every configuration scored against a high-precision reference and priced at cloud rates: faster than every open-source implementation tested, by orders of magnitude in most cases, at equal or better accuracy. Open source (MIT) at github.com/saulpingerman/FastSAR and on PyPI as fastsar; preprint arXiv:2610.33932, submitted to IEEE TGRS.

VESSEL MOVEMENT · FOUNDATION MODEL

A single model pretrained on public AIS traffic that forecasts a vessel's position as a probability map at any horizon, with small task heads on the frozen model for analysts' questions. Paper under review.

Led by the firm's founders. Code, preprints and drafts are available where their status allows.

PRINCIPALS

Paul G. Singerman, Ph.D.

Led the largest AI/ML research program at Penn State's Applied Research Laboratory, a Navy University Affiliated Research Center, as an Associate Research Professor: more than thirty engineers, over ten million dollars a year, and an IDIQ contract with a ceiling in the hundreds of millions of dollars. From 2023 the laboratory's on-site trusted agent to an Intelligence Community agency: contributed to its five-year foundation-model roadmap, technical evaluator on five competitive acquisitions and on BAA panels, coordinated an agency-wide AI/ML training curriculum. Research in geospatial foundation models, SAR image formation and data fusion. Ph.D. electrical engineering and M.A. mathematics, Penn State. TS/SCI with CI polygraph.

John "Luke" Och

Led a team of ten systems engineering and technical assistance (SETA) engineers at SAIC supporting early-stage AI/ML initiatives for a national security space program: acquisition strategies, technical requirements for more than ten low-maturity AI/ML capabilities, and the performance metrics and evaluation criteria behind investment decisions. Five years before that as a satellite ground systems engineer, running source selections from RFP development through proposal evaluation and contract negotiation. Regents Bachelor of Arts, West Virginia University (mathematics and communications); SAFe Agilist. TS/SCI with CI polygraph.